



Department of Electronics & Telecommunication Engineering
Academic Year (2025-2026)

Subject: Principles of Communication Engineering

SY – 5

ASSIGNMENT NO: 2. Pulse Modulation [Module-2]

DATE OF ISSUE: 20/02/2026

CO MAPPING TO THIS
ASSIGNMENT:

CO2: Explain the sampling process and theorem for low-pass signals. Sketch the frequency spectrum for ideal, natural, and flat-top sampling. Draw and describe PCM, DM, and ADM modulators and demodulators.

Q. No	Statement	Marks
1.	State sampling theorem in time domain. Explain sampling process with block diagram.	5M
2.	Explain Aliasing effect & draw the sampled output for sampling frequency less than equal to and greater than maximum frequency of analog signal	5M
3.	State types of quantization. Explain uniform quantization with neat waveform.	5M
4.	Draw & explain spectrum showing aliasing effect & Guard band	5M
5.	Consider the signal $\{3\cos(200\pi t) + (5\sin 6000\pi t) + 10 \cos 1200\pi t\}$ what is Nyquist rate for this signal?	5M
6.	The signal $g(t) = 10 \cos(40\pi t) \cos(400\pi t)$ is sampled at a rate of 500 samples/sec. i) Determine the Nyquist rate ii) Calculate the cut-off frequency of ideal reconstruction filter. iii) Draw the spectrum of resulting sampled signal.	5M
7.	Explain types of sampling with waveform.	5M
8.	What is aliasing? How to reduce it.	5M
9.	Find the Nyquist rate and Nyquist interval for following signal, [5] $X(t) = 3\cos(200\pi t) + 5\sin(600\pi t) + 10\cos(1200\pi t)$	5M
10.	Explain with block diagram and waveform, generation of PAM	5M
11.	Describe generation of flat top samples with circuit diagram & waveform.	5M
12.	What is aperture effect? How to reduce aperture effect.	5M
13.	Define term quantization error. State types of quantization & explain uniform quantization with its characteristics.	5M
14.	State and explain the sampling theorem in detail when $f_s > 2f_m$, $f_s = 2f_m$, $f_s < 2f_m$.	5M



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15.	Explain Flat-top sampling with waveforms.	6M
16.	Distinguish between Ideal sampling, Natural sampling and Flat-Top sampling.	6M
17.	Discuss with block schematic, transmitter & receiver for DPCM (Differential pulse code modulation).	6M
19.	Draw block diagram of Delta modulation system & comment on drawback of Delta modulation.	6M
20.	Explain generation & reconstruction of PCM	6M
21.	Describe Adaptive delta modulation technique & state its advantages	6M
22.	Compare PCM with DM	5M
23.	Describe companding methods μ -law and A - law.	5M
24.	Draw and explain PCM Receiver	5M
24.	Distinguish between DM and ADM.	5M
25.	Explain in detail distortion in delta Modulation.	5M
26.	Define Time Division multiplexing. Explain concept of TDM with neat diagram.	6M
27.	Draw block diagram of PCM system & Describe working of PCM Transmitter.	6M
28.	Discuss with neat schematic, transmitter and receiver for DPCM (Differential pulse code modulation).	6M
29.	Draw Block diagram of Delta modulation system & comment on drawback of Delta Modulation	6M
30.	Explain working of Adaptive Delta Modulation with block diagram & state advantages of ADM over DM	6M
31	Draw and explain slope overload and granular error in DM.	6M
32.	For a 8-bit analog-to-digital converter (ADC) with an input range of 0 V to 5 V, the step size is approximately 1.22 mV and the maximum quantization error is approximately 0.61 mV.	6M
33	Differentiate between A Law and μ Law compandor	6M
34.	Draw and explain with block diagram PCM generation	6M
35.	Compare PCM and Delta Modulation	6M